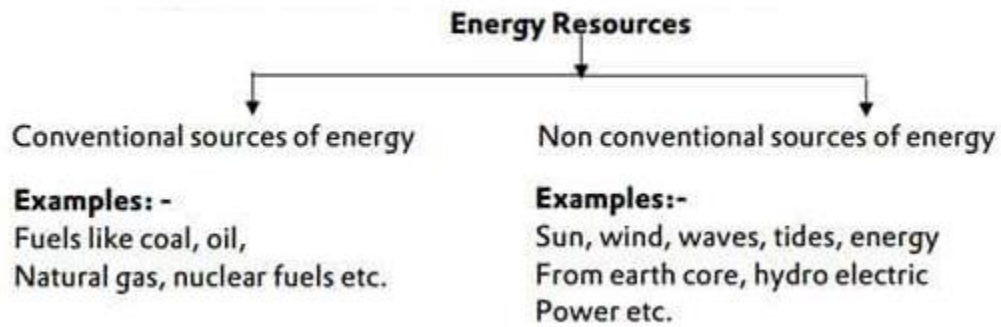


UNIT-III

ENERGY RESOURCES, ELECTRICITY BILL & SAFETY MEASURES

Energy resources can be broadly classified as follows



Difference between conventional and non-conventional sources

	CONVENTIONAL SOURCES OF ENERGY	NON-CONVENTIONAL SOURCES OF ENERGY
1	Conventional sources of energy are the sources that are commonly in use since long time.	Non-conventional sources of energy refers to the sources that are identified few decades ago.
2	They can be exhausted due to over consumption.	They cannot be exhausted.
3	They pollute environment, on a large scale and adds to global warming.	They are environment friendly sources, which does not causes pollution.
4	They are primarily used for industrial and commercial purposes.	They are mainly used for domestic purposes.
5	Costly.	Comparatively less expensive.

Hydro Electric Power Station

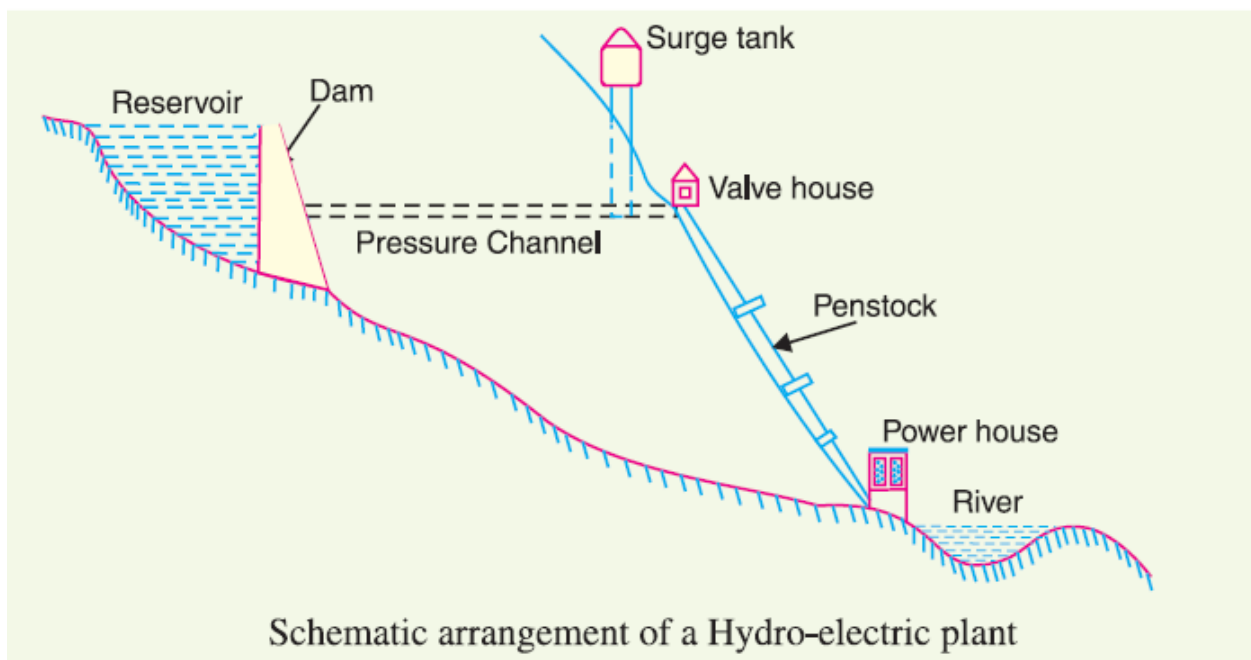
Hydro-electric Power Station

A generating station which utilises the potential energy of water at a high level for the generation of electrical energy is known as a **hydro-electric power station**.

Schematic Arrangement of Hydro-electric Power Station

A hydro-electric power station simply involves the conversion of hydraulic energy into electrical energy. The schematic arrangement of a modern hydro-electric plant is shown in Fig.

The dam is constructed across a river or lake and water from the catchment area collects at the back of the dam to form a reservoir. A pressure tunnel is taken off from the reservoir and water brought to the valve house at the start of the penstock. The valve house contains main sluice valves and automatic isolating valves. The former controls the water flow to the power house and the latter cuts off supply of water when the penstock bursts. From the valve house, water is taken to water turbine through a huge steel pipe known as *penstock*. The water turbine converts hydraulic energy into mechanical energy. The turbine drives the alternator which converts mechanical energy into electrical energy.



A surge tank (open from top) is built just before the valve house and protects the penstock from bursting in case the turbine gates suddenly close due to electrical load being thrown off. When the gates close, there is a sudden stopping of water at the lower end of the penstock and consequently the penstock can burst like a paper log. The surge tank absorbs this pressure swing by increase in its level of water.

What is purpose of surge tank?

Ans. Surge tank is built just before the valve house and it acts as temporary storage reservoir. It is used to suppress the water hammer effect in the penstock.

What is water hammer effect?

A sudden increase in load on the alternator will develop a -ve pressure on the penstock and a sudden removal of load will develop a + ve pressure in the penstock which is known as water hammer effect in the penstock.

Nuclear power Station

Nuclear Power Station

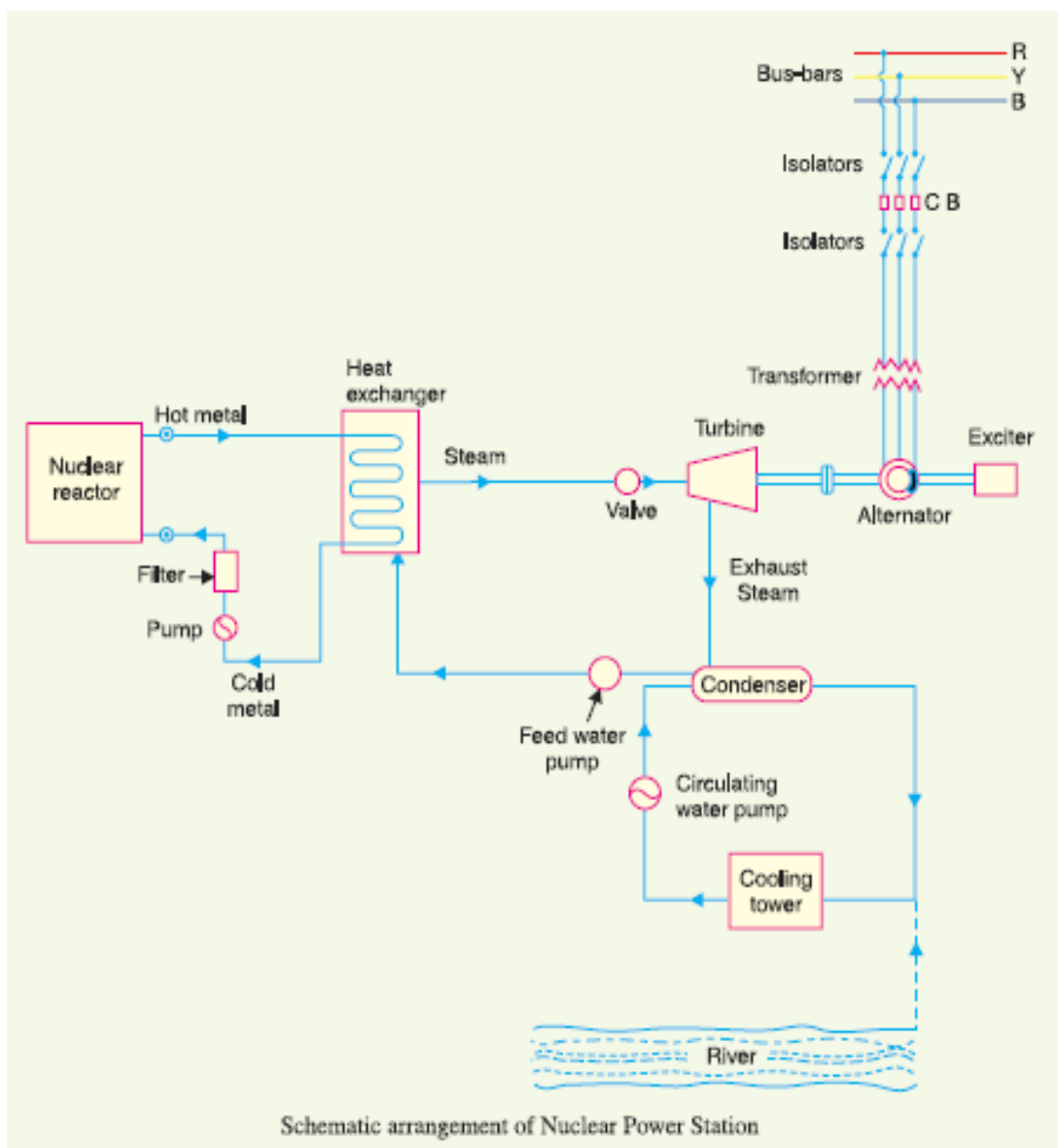
*A generating station in which nuclear energy is converted into electrical energy is known as a **nuclear power station**.*

In nuclear power station, heavy elements such as Uranium (U^{235}) or Thorium (Th^{232}) are subjected to nuclear fission* in a special apparatus known as a *reactor*. The heat energy thus released is utilised in raising steam at high temperature and pressure. The steam runs the steam turbine which converts steam energy into mechanical energy. The turbine drives the alternator which converts mechanical energy into electrical energy.

Schematic Arrangement of Nuclear Power Station

The schematic arrangement of a nuclear power station is shown in Fig. 2.7. The whole arrangement can be divided into the following main stages :

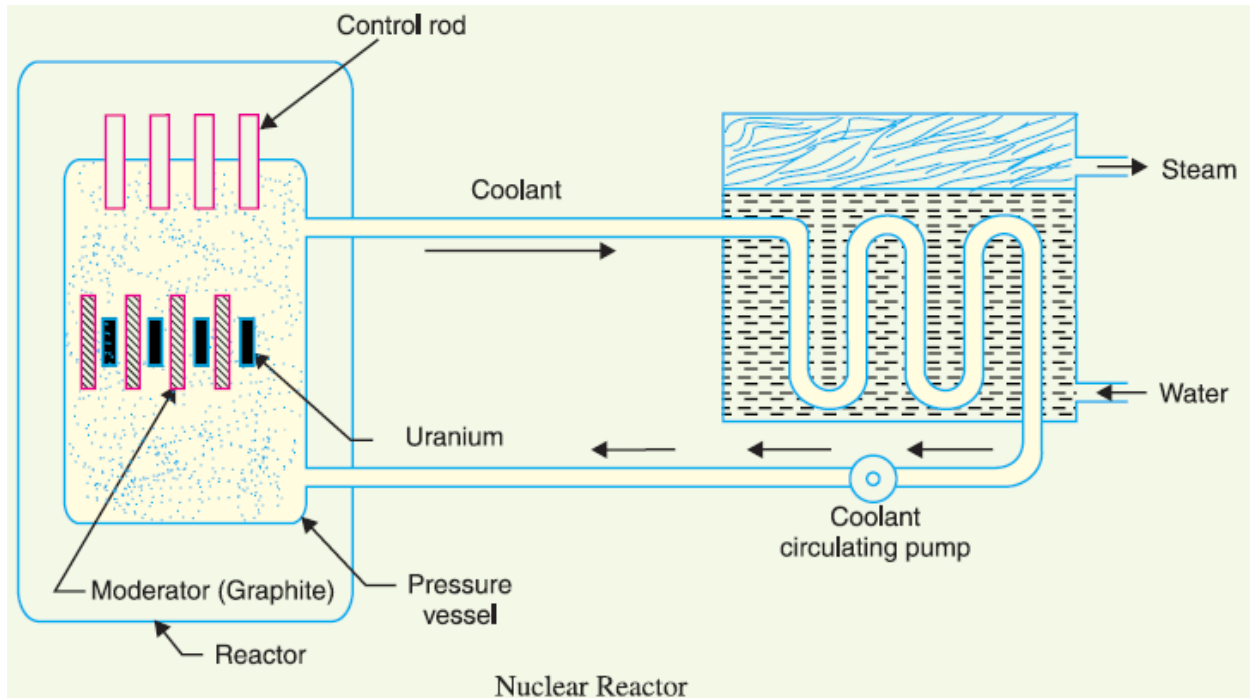
- (i) Nuclear reactor (ii) Heat exchanger (iii) Steam turbine (iv) Alternator.



- (i) **Nuclear reactor.** It is an apparatus in which nuclear fuel (U^{235}) is subjected to nuclear fission. It controls the *chain reaction** that starts once the fission is done. If the chain reaction is not controlled, the result will be an explosion due to the fast increase in the energy released.

A nuclear reactor is a cylindrical stout pressure vessel and houses fuel rods of Uranium, moderator and control rods (See Fig. 2.8). The fuel rods constitute the fission material and release huge amount of energy when bombarded with slow moving neutrons. The moderator consists of graphite rods which enclose the fuel rods. The moderator slows down the neutrons before they bombard the fuel rods. The control rods are of cadmium and are inserted into the reactor. Cadmium is strong neutron absorber and thus regulates the supply of neutrons for fission. When the control rods are pushed in deep enough, they absorb most of fission neutrons and hence few are available for chain reaction which, therefore, stops.

However, as they are being withdrawn, more and more of these fission neutrons cause fission and hence the *intensity* of chain reaction (or heat produced) is increased. Therefore, by pulling out the control rods, power of the nuclear reactor is increased, whereas by pushing them in, it is reduced. In actual practice, the lowering or raising of control rods is accomplished automatically according to the requirement of load. The heat produced in the reactor is removed by the coolant, generally a sodium metal. The coolant carries the heat to the heat exchanger.



- (ii) **Heat exchanger.** The coolant gives up heat to the heat exchanger which is utilised in raising the steam. After giving up heat, the coolant is again fed to the reactor.
- (iii) **Steam turbine.** The steam produced in the heat exchanger is led to the steam turbine through a valve. After doing a useful work in the turbine, the steam is exhausted to condenser. The condenser condenses the steam which is fed to the heat exchanger through feed water pump.
- (iv) **Alternator.** The steam turbine drives the alternator which converts mechanical energy into electrical energy. The output from the alternator is delivered to the bus-bars through transformer, circuit breakers and isolators.

Solar Power Plant

Solar Power Plant:

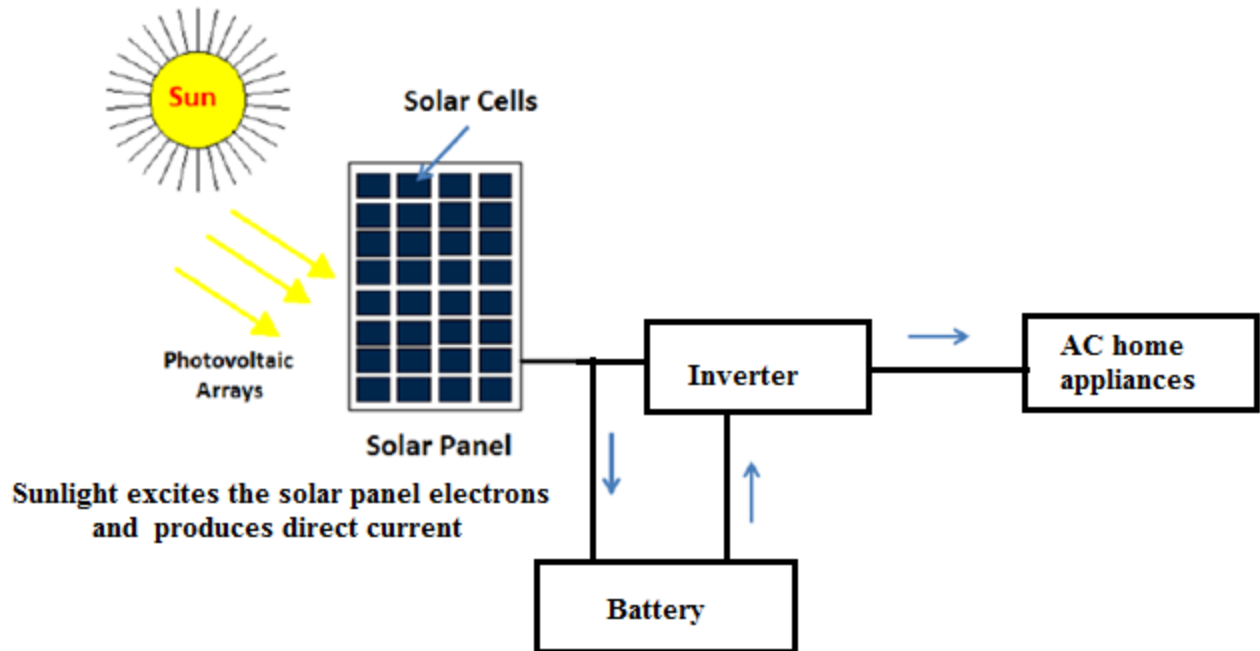


Fig: Working of Solar Power Plant

1. Solar Panels

It is the heart of the solar power plant. Solar panels consist of a number of solar cells. We have got around 35 solar cells in one panel. The energy produced by each solar cell is very small, but combining the energy of 35 of them we have got enough energy to charge a 12 volt battery.

2. Solar Cells

It is the energy generating unit, made up of p-type and n-type silicon semiconductor. It's the heart of solar power plant.

3. Battery

Batteries are used to produce the power back or store the excess energy produced during day, to be supplied during night.

4. D.C. to A.C. Converter (Inverter)

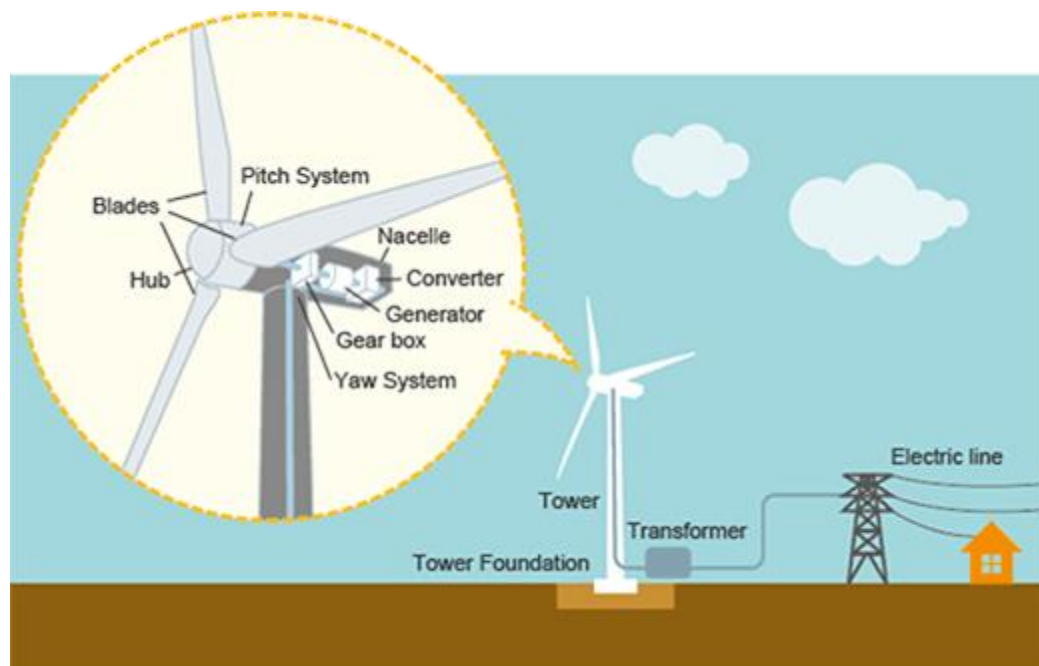
Solar panels produce direct current which is required to be converted into alternating current to be supplied to homes or power grid.

Operation of solar power plant:

As sunlight falls over a solar cells, a large number of photons strike the p-type region of silicon. Electron and hole pair will get separated after absorbing the energy of photon. The electron travels from p-type region to n-type region due to the action of electric field at p-n junction. Further the diode is reversed biased to increase this electric field. So this current starts flowing in the circuit for individual solar cell. We combine the current of all the solar cells of a solar panel, to get a significant output. Solar power plants have a large number of solar panels connected to each other to get a large voltage output. The electrical energy coming from the combined effort of solar panels is stored in the Lithium ion batteries to be supplied at night time, when there is no sunlight.

WIND POWER PLANT

Wind power generation means getting the electrical energy by converting wind energy into rotating energy of the blades and converting that rotating energy into electrical energy by the generator.



How a Wind Turbine Works:

A wind turbine turns wind energy into electricity using the aerodynamic force from the rotor blades, which work like an airplane wing or helicopter rotor blade. When wind flows across the blade, the air pressure on one side of the blade decreases. The difference in air pressure across the two sides of the blade creates both lift and drag. The force of the lift is stronger than the drag and this causes the rotor to spin. The rotor connects to the generator, either directly (if it's a direct drive turbine) or through a shaft and a series of gears (a gearbox) that speed up the rotation and allow for a physically smaller generator. This translation of aerodynamic force to rotation of a generator creates electricity.

How a Wind Plant Works:

Wind power plants produce electricity by having an array of wind turbines in the same location. The placement of a wind power plant is impacted by factors such as wind conditions, the surrounding terrain, access to electric transmission, and other siting considerations. In a utility-scale wind plant, each turbine generates electricity which runs to a substation where it then transfers to the grid where it powers our communities.

Transmission

Transmission lines carry electricity at high voltages over long distances from wind turbines and other energy generators to areas where that energy is needed.

Transformers

Transformers receive AC (alternating current) electricity at one voltage and increase or decrease the voltage to deliver the electricity as needed. A wind power plant will use a step-up transformer to increase the voltage (thus reducing the required current), which decreases the power losses that happen when transmitting large amounts of current over long distances with transmission lines. When electricity reaches a community, transformers reduce the voltage to make it safe and useable by buildings and homes in that community.

Substation

A substation links the transmission system to the distribution system that delivers electricity to the community. Within the substation, transformers convert electricity from high voltages to lower voltages which can then be delivered safely to electricity consumers.

Blades

Most turbines have three blades which are made mostly of fiberglass. Turbine blades vary in size, but a typical modern land-based wind turbine has blades of over 170 feet (52 meters). When wind flows across the blade, the air pressure on one side of the blade decreases. The difference in air pressure across the two sides of the blade creates both lift and drag. The force of the lift is stronger than the drag and this causes the rotor to spin.

Generator

The generator is driven by the high-speed shaft. Copper windings turn through a magnetic field in the generator to produce electricity. Some generators are driven by gearboxes (shown here) and others are direct-drives where the rotor attaches directly to the generator.